

HOMEWORK 3
due Wed, Sep. 23, 11:59 PM
(Upload your solutions to Gradescope)

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1. (4 points) (#28 p. 37.) Find the best-fit least squares parabola to the following data: $\begin{array}{c|cccc} x & 0 & 1 & 3 & 4 \\ \hline y & 0 & 8 & 8 & 20 \end{array}$
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2. (3 points) (#6 p. 35.) For $n > 0$, let

$$f_n(t) = \begin{cases} 1, & 0 \leq t \leq 1/n, \\ 0, & \text{otherwise.} \end{cases}$$

Show that $f_n \rightarrow 0$ in $L^2[0, 1]$. Show that f_n does *not* converge to zero uniformly on $[0, 1]$.

3. (3 points) (#7 p. 35.) For $n > 0$, let

$$f_n(t) = \begin{cases} \sqrt{n}, & 0 \leq t \leq 1/n^2, \\ 0, & \text{otherwise.} \end{cases}$$

Show that $f_n \rightarrow 0$ in $L^2[0, 1]$ but that $f_n(0)$ does not converge to zero.

4. (3 points) (#8 p. 35.) Is Theorem 0.10 true on an infinite interval such as $[0, \infty)$? Give a proof or counterexample. (Theorem 0.10 is on page 10: *If a sequence f_n converges uniformly to f as $n \rightarrow \infty$ on a finite interval $a \leq t \leq b$, then this sequence also converges to f in $L^2[a, b]$.*)
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5. (4 points) (#18 p. 36.) Suppose A is an $n \times n$ matrix with complex entries. Show that the following are equivalent:

- (a) The rows of A form an orthonormal basis in $\mathbb{C}^n$.
 - (b) $AA^* = I$ (the identity matrix).
 - (c) $\|Ax\| = \|x\|$ for all vectors $x \in \mathbb{C}^n$.
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6. (3 points) (#9 p. 84.) Expand the function $f(x) = \cos x$ in a Fourier sine series on the interval $0 \leq x \leq \pi$. Note: this means we expand $f(x) = \sum_{k=1}^{\infty} b_k \sin kx$, where $b_k = \frac{2}{\pi} \int_0^{\pi} f(x) \sin(kx) dx$. (See page 50 of the book.)
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